Madhav Kanda

Applied Researcher | LLM Inference Optimization & Efficient Training

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Work Experiences

Capital One **Jun 2026 – Present**
Senior Associate, Applied Research *San Jose, CA*

- Developed **DStill**, a two-stage framework that distills a target-conditioned diffusion drafter's own multi-step refinement into one-step drafting, improving acceptance length by up to **14%** and inference speed by up to **11% over DFlash**.
- Trained masked diffusion language models as speculative drafters for agentic workflows and synthetic data generation, achieving a **2.5× inference speedup** on synthetic data generation workloads.
- Integrated DStill into DSpark, improving acceptance length by **4.5% over baseline DSpark** and demonstrating applicability across drafting architectures; manuscript **under review at NAACL 2027**.

Microsoft Research **May 2025 – Aug 2025**
Research Intern *Redmond, WA*

- Developed **WorkflowPerturb** to evaluate whether automated metrics reliably capture errors in AI-generated workflows. Curated **44,757 perturbed workflows** for controlled evaluation of metric robustness; accepted at **EMNLP 2026**.

Aalto University, Finland **May 2023 – Jul 2023**
Research Assistant *Espoo, Finland*

- Improved Hamiltonian Monte Carlo sampling efficiency by learning nonlinear reparameterizations during warm-up in **BlackJAX**, increasing effective sample size per gradient evaluation by up to **70%** over a KL divergence-based variational reparameterization baseline on selected models.

Indian Institute of Science **May 2022 – Jul 2022**
Research Intern *Bengaluru, India*

- Developed a seq2seq model for multilingual transliteration and phonetic transcription, improving pronunciation of Indian proper nouns in English with **91.11% accuracy**.

Spyne.ai **Feb 2022 – Apr 2022**
Computer Vision Intern *Remote*

- Implemented a **GAN-based virtual try-on method** using segmentation maps and masking for garment transfer.

Education

University of Illinois Urbana-Champaign (UIUC) **Aug 2024 – May 2026**
Master of Science in Computer Science (Thesis Track) *GPA: 4.0/4.0*

Indian Institute of Technology Gandhinagar (IIT GN) **Nov 2020 – Jun 2024**
Bachelor of Technology with Honours in Computer Science and Engineering *GPA: 9.16/10.0*

Publications

DStill: Trajectory Consistency Distillation for Accelerated One-Step Drafting in Block Diffusion Speculative Decoding.
Shreyas Kulkarni, Uttam Gurram, **Madhav Kanda**, ... Kartik Balasubramaniam, Supriyo Chakraborty, Sambit Sahu.
Under review at NAACL 2027.

RefineStat: Efficient Exploration for Probabilistic Program Synthesis.
Madhav Kanda, Shubham Ugare, Sasa Misailovic.
ICLR 2026 ***(Oral)***.

Constructive Distortion: Improving MLLMs with Attention-Guided Image Warping.
Dwip Dalal, Gautam Vashishtha, Utkarsh Mishra, Jeonghwan Kim, **Madhav Kanda**, ... Heng Ji, Unnat Jain.
ICLR 2026.

WorkflowPerturb: Calibrated Stress Tests for Evaluating Multi-Agent Workflow Metrics.
Madhav Kanda, Sharad Agarwal, Rodrigo Fonseca, Alok Gautam Kumbhare, Pedro Las-Casas.
EMNLP 2026 Industry Track.

(denotes equal contribution)*

Compositional Reasoning via Joint Image and Language Decomposition.
Dwip Dalal*, **Madhav Kanda***, Zhenhailong Wang, Heng Ji, Unnat Jain.
EACL 2026.

Grammar-Constrained LLM Generation for Efficient Probabilistic Program Synthesis.
Madhav Kanda, Shubham Ugare, Sasa Misailovic.
*Workshop on Languages for Inference (LAFI), co-located with **POPL 2026**.*

Towards Scalable Identification of Brick Kilns from Satellite Imagery with Active Learning.
Aditi Agarwal*, Suraj Jaiswal*, **Madhav Kanda***, ... Nipun Batra, Sarath Guttikunda.
***NeurIPS 2023** Workshop on Adaptive Experimental Design and Active Learning in the Real World.*

SpiroActive: Active Learning for Efficient Data Acquisition for Spirometry.
Ankita Kumari Jain, Nitish Sharma, **Madhav Kanda**, Nipun Batra.
arXiv preprint, 2024.

Technical Skills

Programming Languages & Libraries	Python, NumPy, JAX, Triton, pandas, scikit-learn
Deep Learning Frameworks	PyTorch, Transformers
Training & Distributed Systems	SpecForge, Speculators (vLLM)
Inference & Serving	vLLM, SGLang

Achievements

1. Awarded the **Gold Medal for Outstanding Performance** at IIT GN.
2. Named to the **Dean's List** in **all semesters** at IIT GN.
3. Selected for the **Indian Academy of Sciences Summer Research Fellowship**.
4. Awarded **Academic Research Ranking 2** in the Computer Science and Engineering discipline at IIT GN for recommendation for exchange programmes.
5. Acknowledged for contributions to *Probabilistic Machine Learning: Advanced Topics* by Kevin P. Murphy.

Academic Service

- **Reviewer:** ICLR 2027, NAACL 2027, NeurIPS 2026, and EMNLP 2026.
- **Student Volunteer, POPL 2026:** Supported tutorials, registration, and conference logistics.
- Acknowledged benchmark contributor, **ScholarCatalyst: A Benchmark for Retrieving Papers That Inspire New Research** (2026).

Talks & Tutorials

1. Evaluating Metrics for Agentic Workflows → **Microsoft Research**
2. RefineStat: Efficient Exploration for Probabilistic Program Synthesis → **ICLR'26**
3. Tutorial on Variationally Inferred Parameterization → **NumPyro**

Teaching

- Teaching Assistant, **CS 225: Data Structures**, UIUC (three semesters, 2024–2026).
- Teaching Assistant, **ES 301: Data Structures and Algorithms-II**, IIT Gandhinagar (Spring 2024).
- Teaching Assistant, **ES 661: Probabilistic Machine Learning**, IIT Gandhinagar (Fall 2023).
- Teaching Assistant, **Bayesian Data Analysis**, Aalto University (Spring 2023; online).